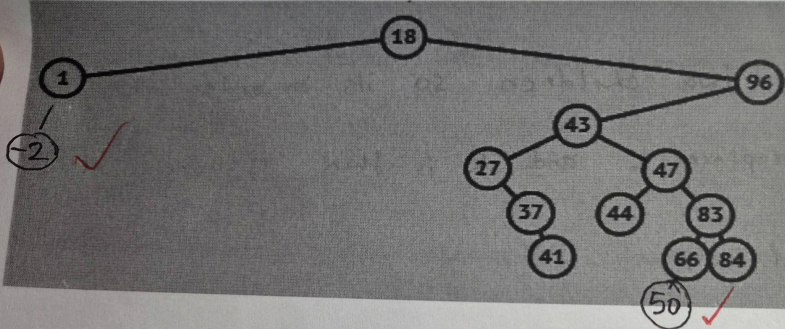


Q5. Given the following Binary Search Tree (BST), answer the following questions. [6

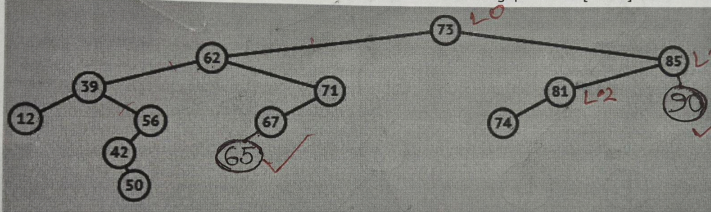


1. Insert 50 in this BST (Show it on the given image without redrawing.) [mark-1]
2. Insert -2 in this BST (Show it on the given image without redrawing.) [mark-1]
3. Height of the tree is 5 [1 mark] ✓
4. Height of node 27 is 2 [1 mark] ✓
5. The level of node 27 is 3 [1 marks] ✓
6. The degree of node 27 is 1 [1 marks] ✓
7. Perform Post-order traversal on the original tree (without insertion). [2 marks]

1, 41, 37, 27, 44, 66, 84, 83, 47, 43, 96, 18 ✓

Post-order: 1, 41, 37, 27, 44, 66, 84, 83, 47, 43, 96, 18.

Q5. Given the following Binary Search Tree (BST), answer the following questions. [6+2=8]



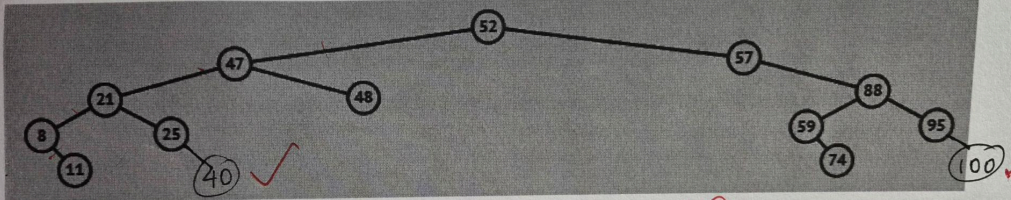
1. Insert 65 in this BST (Show it on the given image without redrawing.) [mark-1]
2. Insert 90 in this BST (Show it on the given image without redrawing.) [mark-1]
3. Height of the tree is 5 [1 mark] ✓
4. Height of node 56 is 2 [1 mark] ✓
5. The level of node 81 is 2 [1 marks] ✓
6. The degree of node 81 is 1 [1 marks] ✓

7. Perform Post-order traversal on the original tree (without insertion). [2 marks]

12 → 50 → 42 → 56 → 39 → 67 → 71 → 62 → 74 → 81 → 90 → 85 → 73 ✓

Post-order: 12, 50, 42, 56, 39, 67, 71, 62, 74, 81, 90, 85, 73.

Q5. Given the following Binary Search Tree (BST), answer the following questions. [6+2=8]



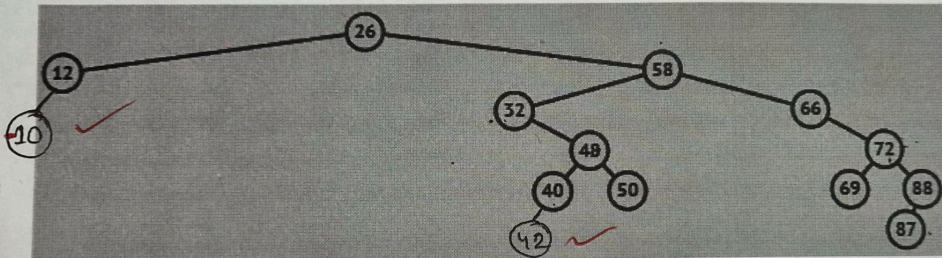
1. Insert 40 in this BST (Show it on the given image without redrawing.) [mark-1] ✓
2. Insert 100 in this BST (Show it on the given image without redrawing.) [mark-1] ✓
3. Height of the tree is 4 [1 mark] ✓
4. Height of node 57 is 1 [1 mark] ✓
5. The level of node 21 is 2 [1 marks] ✓
6. The degree of node 21 is 2 [1 marks] ✓

7. Perform Post-order traversal on the original tree (without insertion). [2 marks]

11, 8, 40, 25, 21, 48, 47, 74, 59, 100, 95, 88, 57, 52 ✓

Post-order: 11, 8, 25, 21, 48, 47, 74, 59, 95, 88, 57, 52.

Q5. Given the following Binary Search Tree (BST), answer the following questions. [6+2=8]



1. Insert 42 in this BST (Show it on the given image without redrawing.) [1 mark]
2. Insert -10 in this BST (Show it on the given image without redrawing.) [1 mark]
3. Height of the tree is 5 [1 mark] ✓
4. Height of node 72 is 2 [1 mark] ✓
5. The level of node 48 is 3 [1 marks] ✓
6. The degree of node 48 is 2 [1 marks] ✓

7. Perform Post-order traversal on the original tree (without insertion). [2 marks]

12, 40, 50, 48, 32, 69, 87, 88, 72, 66, 58, 26 ✓

Post-order: 12, 40, 50, 48, 32, 69, 87, 88, 72, 66, 58, 26.

=====

Q1. In a Full (Strict) Binary Tree, a node can have only 0, 1, or 2 children. [1 mark] [True/False]

Solution: False

Explanation: In a full binary tree, every node has either 0 or 2 children, never exactly 1 child.

Q2. In a BST, a new node is always inserted as a leaf node. [1 mark] [True/False]

Solution: True

Explanation: A new node is inserted at the first available null position after following the BST ordering rules.

Q3. In a BST, a new node is always inserted as the root node. [1 mark] [True/False]

Solution: False

Explanation: A new node becomes the root only when the BST is empty.

Q4. In a balanced BST, the time complexity of inserting one element is _____. [1 mark]

Solution: $O(\log n)$

Explanation: A balanced BST has height $O(\log n)$, so insertion requires traversing at most that height.

Q5. In a skewed BST, the time complexity of inserting one element is _____. [1 mark]

Solution: $O(n)$

Explanation: A skewed BST may have height $n-1$, so insertion can require traversing almost every node.

Q6. In a skewed BST, the time complexity for searching one element is _____. [1 mark]

Solution: $O(n)$

Explanation: A skewed BST behaves like a linked list, so searching may require checking all nodes.

Q7. For a tree (or free tree) with n nodes, the number of edges is equal to: _____

Solution: $n-1$

Explanation: Every tree with n nodes contains exactly $n-1$ edges.

Q8. In a Complete Binary Tree, if an internal node has only one child, it must be the right child. [True/False]

Solution: False

Explanation: In a complete binary tree, a node with only one child must have a left child.

Q9: In a Perfect Binary Tree, all leaf nodes are at the same level. [Mark-1] [True/False]

Solution: True

Explanation: By definition, all leaf nodes in a Perfect Binary Tree are at the same level.

Q10: In a Perfect Binary Tree, every internal node has exactly two children. [Mark-1] [True/False]

Solution: True

Explanation: Every non-leaf (internal) node in a Perfect Binary Tree has exactly two children.
